

10/30 Lectures 10 & 11

Last time: * Defn of uniformly hyperbolic flows, invariant manifolds & subspaces

Now: stable / unstable manifold theorem

Note: although we prove this only for hyperbolic fixed pts, versions of the theorem for hyperbolic maps / flows.

A hyperbolic map is a discrete-time dynamical system in which at every point x , you have a stable-unstable decomposition of the tangent space.

$$\text{i.e. } T_x \mathbb{R}^n = E^u(x) \oplus E^s(x)$$

(which you should think of as "velocity" or inf. perturbations in $\mathbb{R}^n$ at x)

$$\text{Map: } F: \mathbb{R}^n \rightarrow \mathbb{R}^n$$

$$F(x_t) = x_{t+1}$$

$$F^t = \underbrace{F \circ \dots \circ F}_{t \text{ times}} = F \circ F^{t-1}$$

$$\text{with } E^u(x) = \{ v \in T_x \mathbb{R}^n : \|\nabla F^t(x) v\| \leq C e^{-\alpha t} \|v\|, t \rightarrow -\infty \}$$

$$E^s(x) = \{ v \in T_x \mathbb{R}^n : \|\nabla F^t(x) v\| \leq C e^{-\alpha t} \|v\|, t \rightarrow \infty \}$$

These constants $\alpha, C > 0$ are again uniform in x .

More precisely, such an F is called a uniformly hyperbolic map

E.g. a linear hyperbolic map

$$F(x) = Ax \text{ satisfies the definition}$$

above when A has no eigenvalue with absolute value 1.

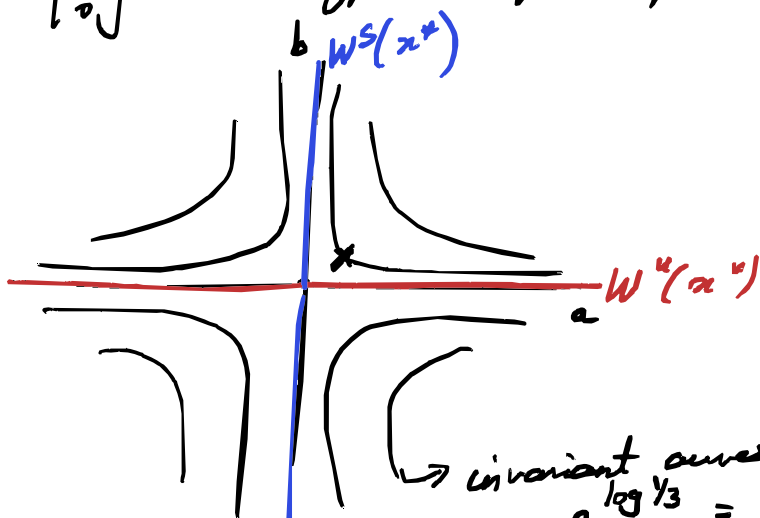
E^s at any x is the stable generalized eigenspace of A . i.e. generalized eigenspace for eigenvalues with absolute value < 1

Similarly, E^u at any x is the generalized eigenspace for eigenvalues with absolute value > 1 .

$$F(x) = \begin{bmatrix} 2 & 0 \\ 0 & 1/3 \end{bmatrix} x \quad x \in \mathbb{R}^2.$$

Ex: check that E^s & E^u are the standard coordinate directions.

$x = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ is a hyperbolic fixed point



Ex.

Take a point such as one marked x .

What are its W^u & W^s ?

Stable manifolds & unstable manifolds ^{and subspaces} swap when we reverse time, i.e. for the inverse dynamics

So WLOH, we prove only the stable manifold theorem but analogous statements hold for the unstable manifold.

Local Stable manifold

$$W^{s,loc}(x^*) = \{ y \in W^s(x) : \|\varphi^t(y) - x^*\| \leq C e^{-\alpha t}, \forall t \geq 0 \}$$

The stable manifold theorem says that a local stable manifold around x^* exists & is C^k if the flow is locally C^k . Further, $W^{s,loc}$ is a Lipschitz graph over $E^s(x^*)$ and is tangent to $E^s(x^*)$ at x^* .

Things to know for the proof:

- Contraction mapping principle on complete metric spaces (here used on a Banach space, ^{factored} subset of continuous functions of time into $\mathbb{R}^n$). Contraction is used to justify the exists of points that remain in a small neighborhood of x^* . Size of nbd appears in the definition of the Banach space
- The fixed point of the contraction above is the orbit of a point on W_{loc}^s . To show this orbit converges exponentially to the fixed point, a generalized Grönwall's inequality is used. Fixed point is a function of $E^s \Rightarrow W_{loc}^s$ is a graph over E^s
- To show it is a Lipschitz graph ^{over E^s} & so are its unstable projections, again a generalized Grönwall's inequality is used.
- The Contraction map is a "uniform contraction" over E^s . That is, the ^{rate of} contraction are uniform over $E^s \Rightarrow$ fixed point is unique and has the same order of differentiability as the Contraction map

Now in math!

Let x be a point, close to x^* . within the neighb of x^* where the flow is C^k .

$$\text{Let } y(t) = \varphi^t(x) - x^*$$

$$\text{Let } x - x^* = y(0) = (y(0))_s + (y(0))_u$$

$$\text{where } (y(0))_s \in E_s(x^*)$$

We know that

$$\begin{aligned} \frac{\partial}{\partial t} y(t; y(0)) &= \frac{\partial}{\partial t} \varphi^t(x) \\ &= V(y(t; y(0)) + x^*) \\ &= \underbrace{\nabla V(x^*)}_A y(t; y(0)) + N(y(t; y(0))) \end{aligned}$$

Ansatz for solution:

$$\begin{aligned} y(t) &= e^{tA} (y(0))_s + \int_0^t e^{-(t-\tau)A} (N(y(\tau)))_s d\tau \\ &\quad - \int_t^\infty e^{-(t-\tau)A} (N(y(\tau)))_u d\tau \end{aligned}$$

Ex: check this ansatz satisfies the ODE

Note: $y(t; y(0))$ is written in terms of the stable component of the initial comp. $(y(0))_s$.

Let's call this $s = (y(0))_s$

here A is linear part & $N(y)$ is nonlinear in y .

It turns out if $\|y(t)\|$ is bounded, we can write a solution to the ODE

$$\frac{dy}{dt} = Ay + N(y)$$

as

$$y(t) = e^{tA} (y(0))_s + \int_0^t e^{(t-\tau)A} (N(y(\tau)))_s d\tau - \int_t^\infty e^{(t-\tau)A} (N(y(\tau)))_u d\tau$$

To show that this solution is unique if $(y(0))_s$ is in a small nbd of 0, define a map T from the space of continuous functions on $[0, \infty)$ (time) to $\mathbb{R}^n : C^0([0, \infty), \mathbb{R}^n)$ onto itself.

$$T(\varphi) = e^{tA} s + \int_0^t e^{(t-\tau)A} (N(y(\tau)))_s d\tau - \int_t^\infty e^{(t-\tau)A} (N(y(\tau)))_u d\tau$$

↑
function in $C^0([0, \infty), \mathbb{R}^n)$

Here s is some fixed vector in $E^s(x^*)$.

Ex: Show that if $N(y)$ is in C^r on a nbd of 0, so is $T(\varphi)$

Take a closed subset of $C^0([0, \infty), \mathbb{R}^n)$

$$V_\delta = \left\{ t \rightarrow y(t) : \sup_{t \in [0, \infty)} \|y(t)\| \leq \delta \right\}$$

Ex: Show that if $N(y)$ is in C^r whenever $\|y\| < \delta_0$, $\exists$

Now, if $y(t)$ is bdd, this is the only soln.

In general if you take an x in a small neighborhood of x^* , $y(t) = \varphi^t(x) - x^*$ could be unbounded

But if $y(t)$ is bdd, it satisfies

$$y(t) = e^{tA}s + \int_0^t e^{(t-\tau)A} (N(y(\tau)))_s d\tau - \int_t^\infty e^{(t-\tau)A} (N(y(\tau)))_u d\tau$$

(where v_u, v_s means unstable/stable components of $v = v_u + v_s$).

How to prove that y is unique?

(Contraction mapping principle.

Need to write a map s.t. $t \mapsto y(t)$ above is a fixed pt.

Let $C^0([0, \infty), \mathbb{R}^n)$ be the space of continuous fns from $[0, \infty)$ to $\mathbb{R}^n$.

Ex: let y be an element of this space.

$$\text{Show that } T(y; s) := e^{tA}s + \int_0^t e^{(t-\tau)A} (N(y(\tau)))_s d\tau - \int_t^\infty e^{(t-\tau)A} (N(y(\tau)))_u d\tau$$

is well-defined on this space.

That is $T(y)$ is also $\in C^0([0, \infty), \mathbb{R}^n)$

Let $\delta > 0$. We will define a closed subset of $C^0([0, \infty), \mathbb{R}^n)$ and show that $T(\cdot; s)$ is a contraction on this space

$$V_\delta = \left\{ y \in C^0([0, \infty), \mathbb{R}^n) : \|y\| := \sup_{t \in [0, \infty)} \|y(t)\| < \delta \right\}$$

First let's find δ appropriately so that $T(\cdot; s)$ maps V_δ to V_δ .

and show that δ does not depend on s

$$\|T(y, s)\| \leq \|e^{tA}s\| + \int_0^t \|e^{(t-\tau)A} (N(y(\tau)))_s\| d\tau + \int_t^\infty \|e^{(t-\tau)A} (N(y(\tau)))_u\| d\tau$$

Since $\|e^{tA}s\| \leq C e^{-\alpha t} \|s\|$ $s \in E^s(t)$ and $\|e^{tA}u\| \leq C e^{-\alpha(t)} \|u\|$ $u \in E^u(t)$

$$\|T(y, s)\| \leq C \|s\| + C \int_0^\infty e^{-\alpha \tau} \|(N(y(\tau)))_s\| d\tau + C \int_0^\infty e^{-\alpha \tau} \|(N(y(\tau)))_u\| d\tau$$

$N \in C^2$, $\alpha > 1$, in a nbd around 0 by assumption. Thus, $N(v) = o(\|v\|)$, i.e.

For any $\varepsilon > 0$, $\exists \delta > 0$ s.t. $\|N(v)\| < \varepsilon \|v\|$ whenever $\|v\| < \delta$.

Since $(N(v))_s = P_s N(v)$

$$\|(N(v))_s\| \leq \|P_s\| \|N(v)\| \leq \|P_s\| \varepsilon \|v\|$$

Hence, $\|T(y, s)\| \leq C \|s\| +$

$$\|P_s\| \varepsilon C \int_0^\infty e^{-\alpha \tau} \|y(\tau)\| d\tau + \varepsilon C \|P_u\| \int_0^\infty e^{-\alpha \tau} \|y(\tau)\| d\tau$$

Thus, when $\|y(\tau)\| < \delta$ $\forall \tau$,

$$\|T(y, s)\| \leq C \|s\| + \frac{\|P_s\|}{\alpha} \varepsilon C \delta + \frac{\|P_u\|}{\alpha} \varepsilon C \delta$$

So choosing δ & ε so that

$$C \|s\| \leq \frac{\delta}{3}$$

$$C \|P_s\| \varepsilon \frac{\delta}{\alpha} \leq \frac{\delta}{3}$$

$$\text{and } C \varepsilon \|P_u\| \frac{\delta}{\alpha} < \frac{\delta}{3},$$

$$\text{we get if } \delta \geq 3C e^{-\alpha t} \|s\|$$

$$\varepsilon \leq \min \left\{ \frac{\alpha}{3C \|P_s\|}, \frac{\alpha}{3C \|P_u\|} \right\}$$

then, $\|T(y, s)\| \leq \delta$

for s with $\|s\| \leq \delta/3C$

Thus, $T(y, s)$ maps V_δ on to itself

Now we have to show that $T(y, s)$ is a contraction V_δ .

Consider

$$\|T(y, s) - T(y', s)\| \leq \left\| \int_0^t e^{-(t-\tau)A} ((N(y(\tau)))_s - (N(y'(\tau)))_s) d\tau \right\| + \left\| \int_t^\infty e^{-(t-\tau)A} ((N(y(\tau)))_u - (N(y'(\tau)))_u) d\tau \right\|$$

$$\leq \|P_s\| \frac{C\varepsilon}{\alpha} \|y - y'\| + \|P_u\| \frac{C\varepsilon}{\alpha} \|y - y'\|$$

Since $\|P_s\| \frac{C\varepsilon}{\alpha} + \|P_u\| \frac{C\varepsilon}{\alpha} < \frac{2}{3}$

$\Rightarrow T(\cdot, s)$ is a contraction for any s when $\|s\| < \delta/3C$.

From contraction mapping principle, a unique fixed point y exists on V_δ .

This unique fixed point is a function of s . Thus, a graph over E_s .

Let y_s be the fixed pt of $T(\cdot, s)$.

Show that y is a Lipschitz function of s .

$$\|y_s(t) - y_{s'}(t)\| \leq \|s - s'\| + \left\| \int_0^t e^{-(t-\tau)A} ((N(y_s(\tau)))_s - (N(y_{s'}(\tau)))_s) d\tau \right\| + \left\| \int_t^\infty e^{-(t-\tau)A} ((N(y_s(\tau)))_u - (N(y_{s'}(\tau)))_u) d\tau \right\|$$

$$\leq e^{-\alpha t} \|s - s'\| + \|P_s\| C \varepsilon \int_0^t e^{-(t-\tau)\alpha} \|y_s(\tau) - y_{s'}(\tau)\| d\tau + \|P_u\| C \varepsilon \int_t^\infty e^{-(t-\tau)\alpha} \|y_s(\tau) - y_{s'}(\tau)\| d\tau$$

By applying generalized Grönwall's inequality (lemma 5.10 in Meiri)

$$\text{we get } \|y_s(t) - y_{s'}(t)\| \leq C' \|s - s'\| e^{-(\alpha - C\varepsilon)t}$$

for some $\delta_0 > 0$ & $C' > 0$.

$$\Rightarrow \|y_s - y_{s'}\| \leq C' \|s - s'\|$$

Lipschitz const

Repeat the same idea to show that

$$P_u(y_s(t) - y_{s'}(t)) = (y_s(t) - y_{s'}(t))_u$$

is also Lipschitz in s .

$$\Rightarrow \{y(t) \text{ (the fixed pt)} = \varphi^t(x) - x^*\}_t \text{ is a Lipschitz graph over } E^s(x^*).$$

To show the differentiability of y w.r.t s , we can invoke a fact about contraction maps (see Theorem 5.11 in Meiri and also its proof): if the map $T(\cdot, s) \in C^r$ and the contraction rate is uniform over s , then y_s is C^r in s .

To check the conditions of this result recall the definition of $\frac{\partial}{\partial s} A$

$$T(y, s) = e^{tA}s + \int_0^t e^{(t-\tau)A} (N(y(\tau)))_s d\tau - \int_t^\infty e^{(t-\tau)A} (N(y(\tau)))_u d\tau$$

$y \mapsto T(y, s) \in C^1$ if N is C^1 .

$\Rightarrow$ applying theorem, y_s is a C^1 function of s .

$$\text{Since } y_s(t) = e^{tA}s + \int_0^t e^{(t-\tau)A} (N(y_s(\tau)))_s d\tau - \int_t^\infty e^{(t-\tau)A} (N(y_s(\tau)))_u d\tau$$

Differentiating $y_s(t)$ w.r.t s ,

$$\frac{\partial}{\partial s} y_s(t) = e^{tA} + \int_0^t e^{(t-\tau)A} P_s \nabla N(y_s(\tau)) \frac{\partial}{\partial s} y_s(\tau) d\tau - \int_t^\infty e^{(t-\tau)A} P_u \nabla N(y_s(\tau)) \frac{\partial}{\partial s} y_s(\tau) d\tau$$

when $s = 0$, i.e. there is no inf. perturb. along E_s , and if $y_s(t) \in W^{s,loc}/\text{bdd}$

$\Rightarrow y_s(t) = 0$ at all t . (by assumption)

Now, at $x = x^*$, $\nabla N(0) = 0$.

when $y_s = 0$

$$\Rightarrow \frac{\partial}{\partial s} y_s(t) = e^{tA}$$

$$\Rightarrow \text{for any } v \in E^{s,loc}, \frac{\partial}{\partial s} y_s(t) v \in E^s(x^*)$$

i.e. if $y_s(t) \in W^{s,loc}$, at $s = 0$,

$\frac{\partial}{\partial s} y_s(t)$ maps E^s on to itself.

$\Rightarrow W^{s,loc} \cap E^s$ are tangent at $x = x^*$.

Pictorially, W^s_{loc} is a C^r graph

i.e. any point x on W^s_{loc}

can be written in its $E^u \oplus E^s$ coordinates as

$$\left(\frac{\partial}{\partial s} y_s(t), s \right)$$